

Assignment 2: Vlan and Secure Switch Configuration

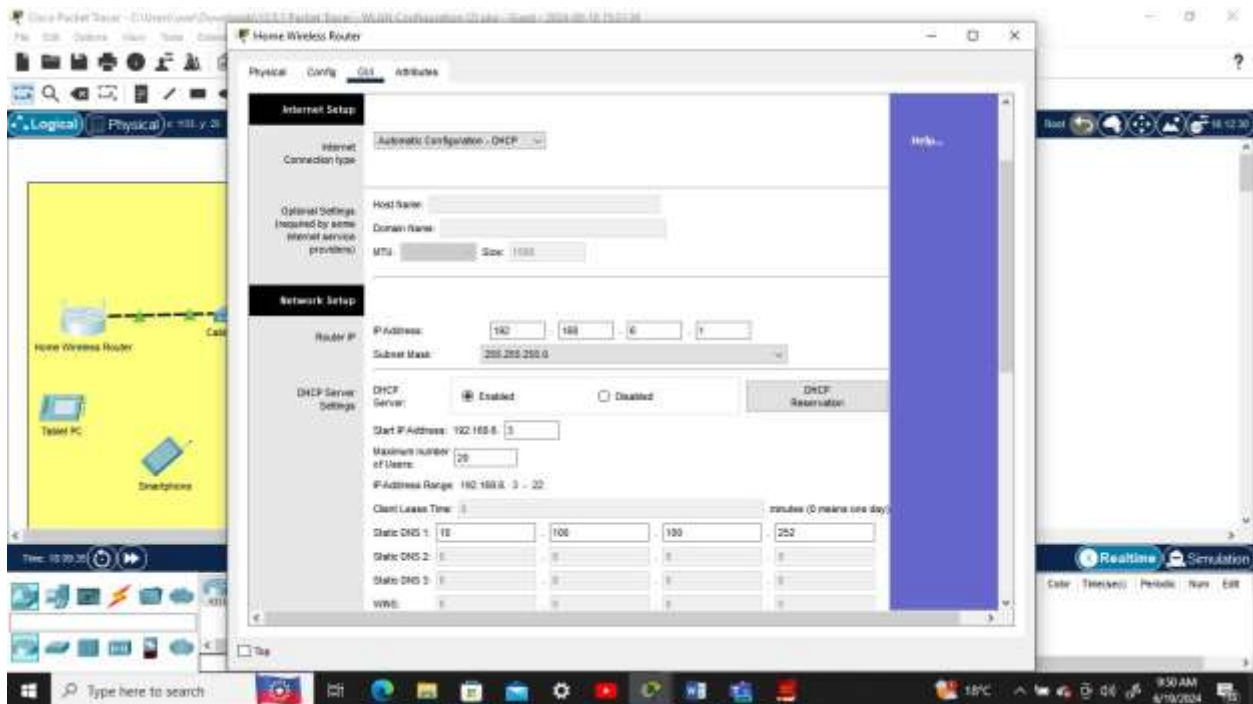
Report by Ezekiel Waweru CNS06-24049

INTRODUCTION

This assignment dealt with configuring a wireless home router and a wireless LAN controller based network. I also implemented Wi-Fi protected access II- Pre-Shared Key (WPA2-PSK) and Wi-Fi protected access- Enterprise (WPA2-Enterprise) security.

Part 1: Configure a Home Wireless Router

Step 1: Change DHCP settings



I opened the Home wireless router GUI and changed the router IP and DHCP settings according to the information in the Addressing Table.

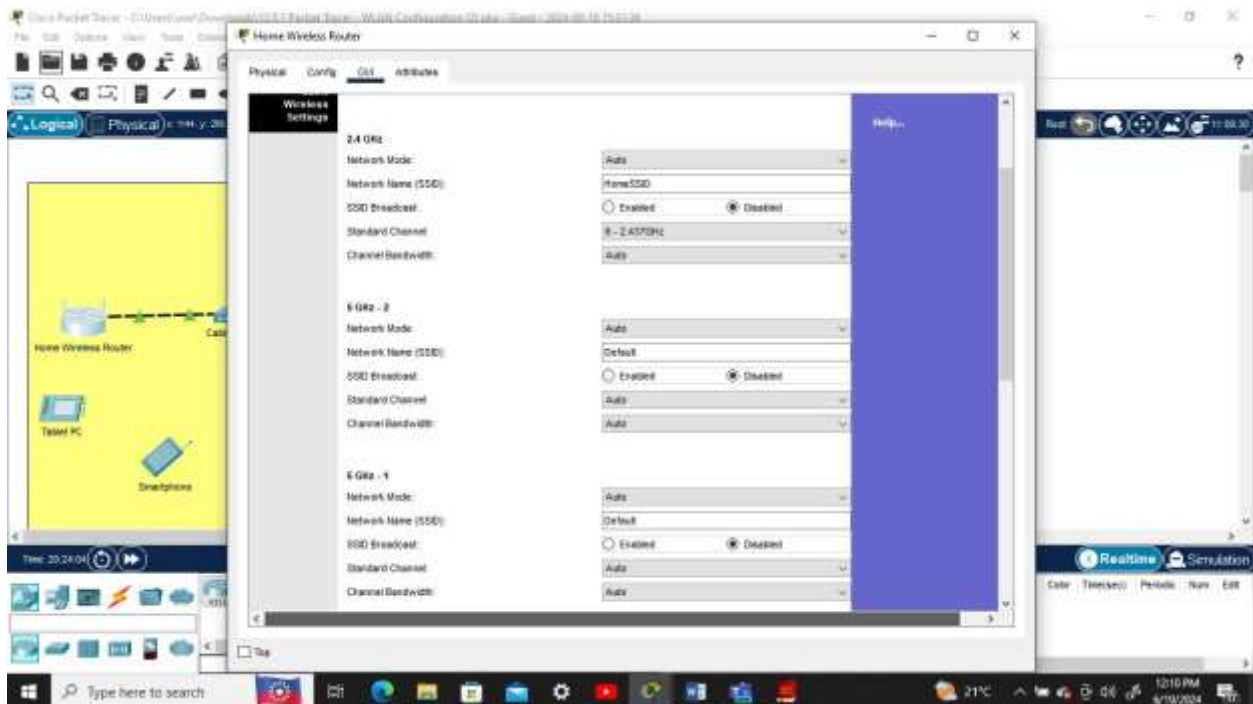
Next, I permitted a minimum of 20 addresses to be issued by the router.

I then configured the DHCP server to start with IP addresses .3 of the LAN network

After, I configured the internet interface of the router to receive its IP address over DHCP. Verifying the address, this is the address it received

I then configured the static DNS server to the address in the Addressing table

Step 2: Configure the wireless LAN

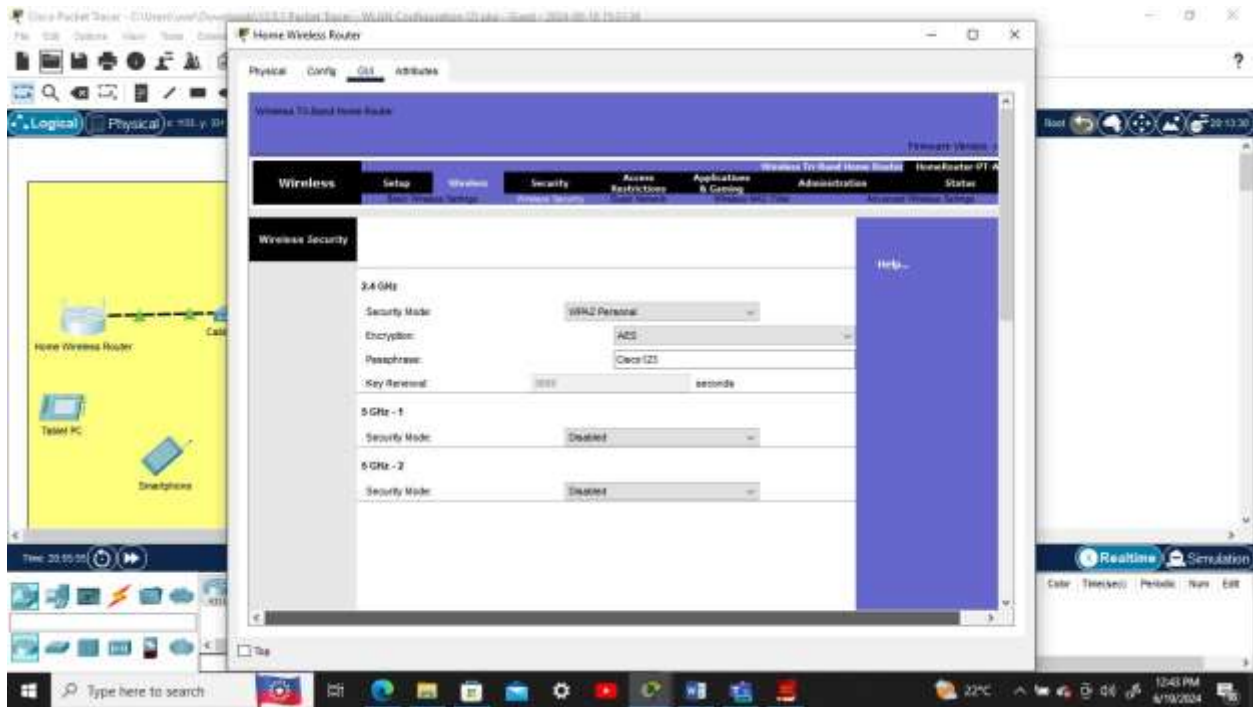


I configured the network with the SSID as was shown in the Wireless LAN table information

The channel used was 6

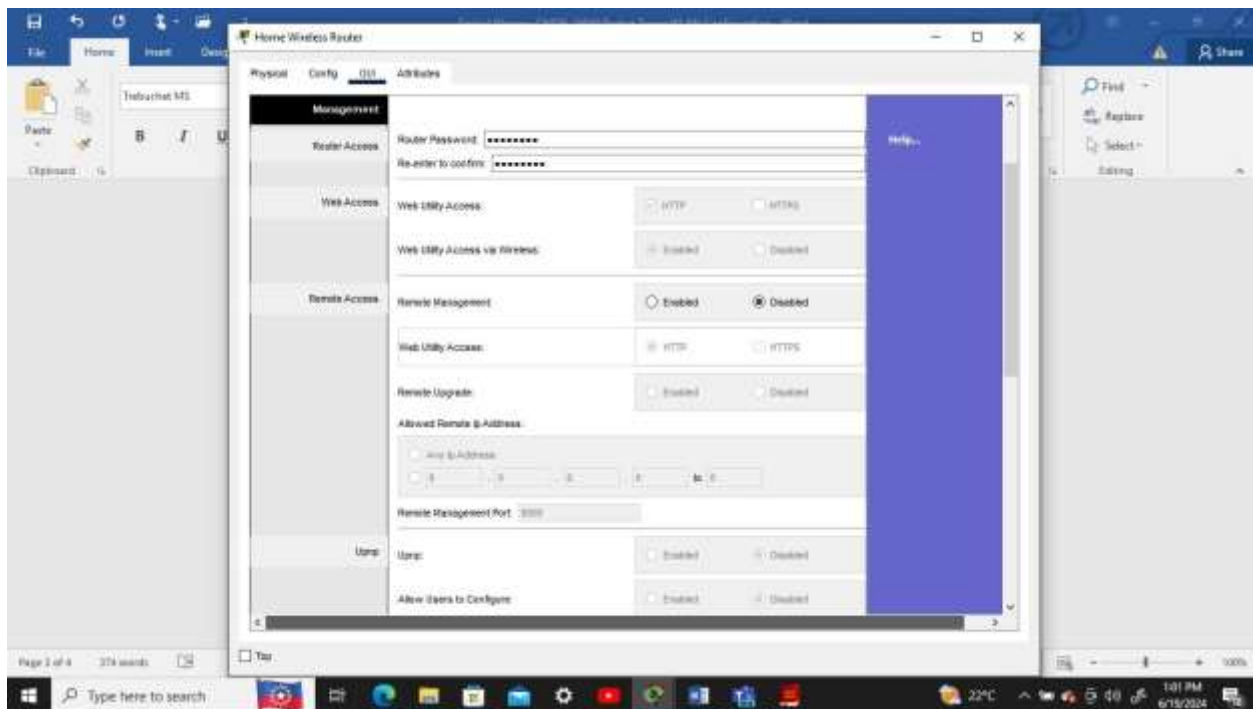
Ensuring that all wireless hosts in the home were able to see the SSID was the last step

Step 3: Configure Security



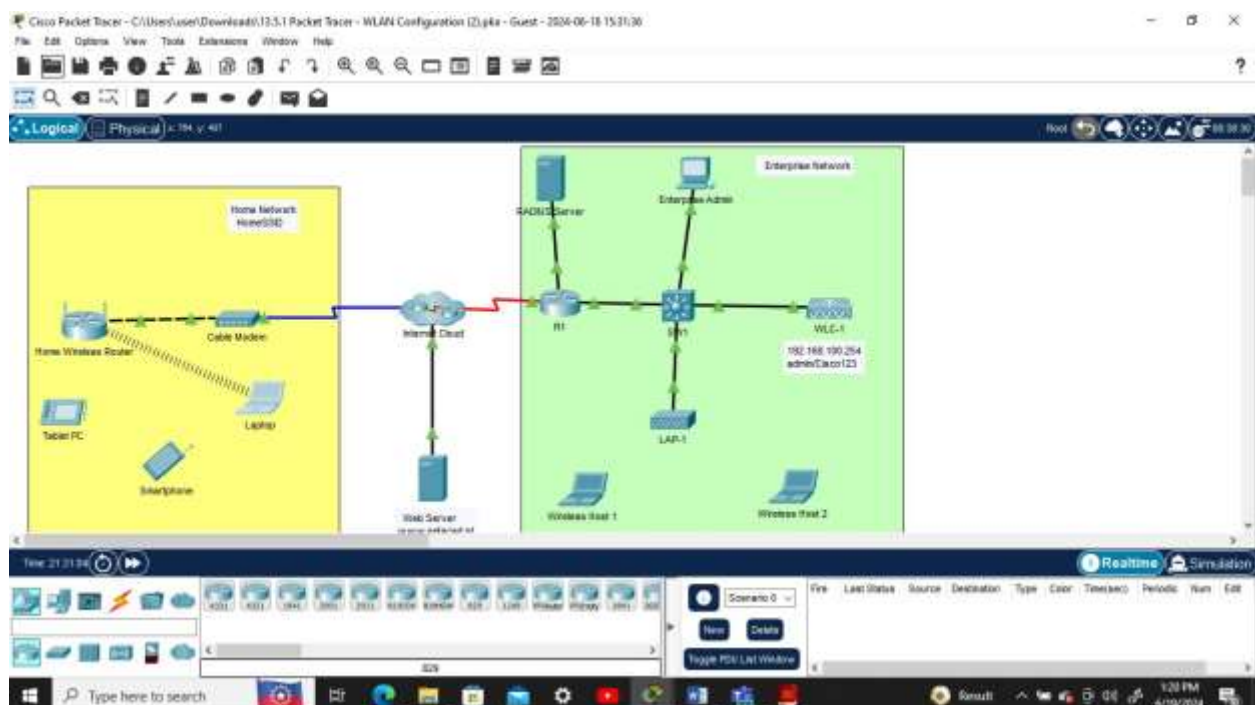
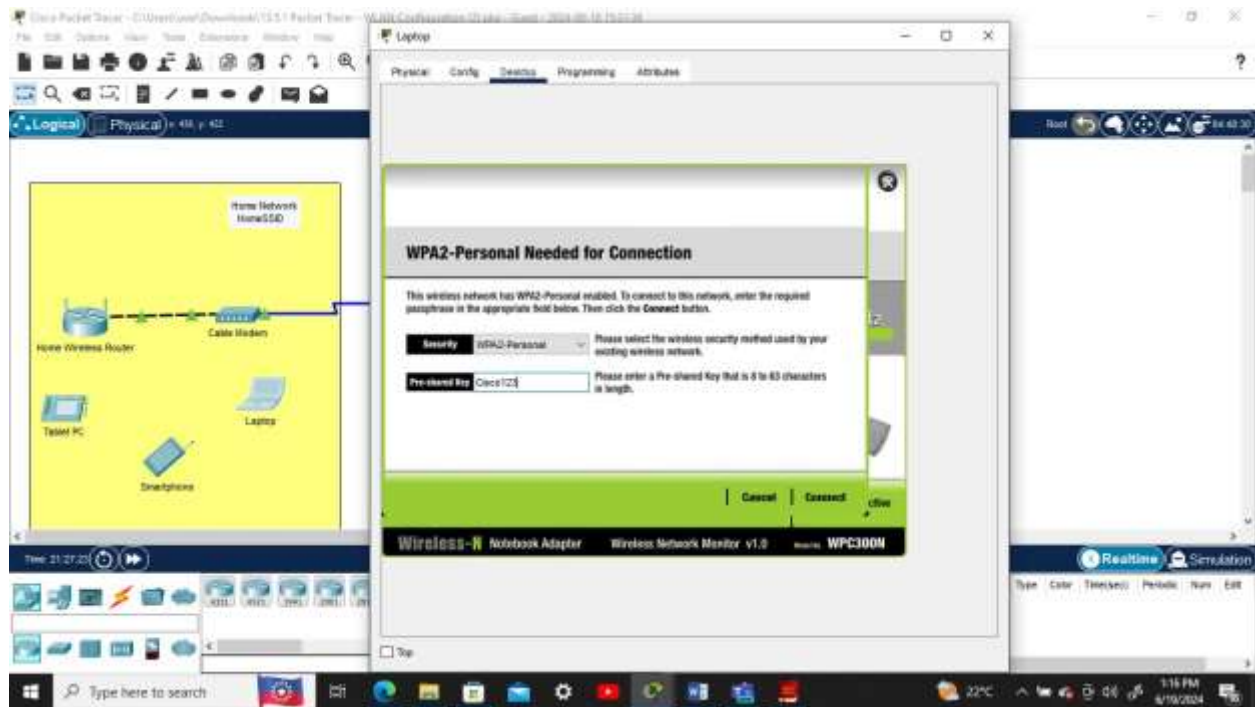
I configured the wireless LAN security and also used the WPA2 Personal and the passphrase shown in the wireless LAN information table

Securing the router by changing the default password to the value shown in the Wireless LAN information table was the next step.

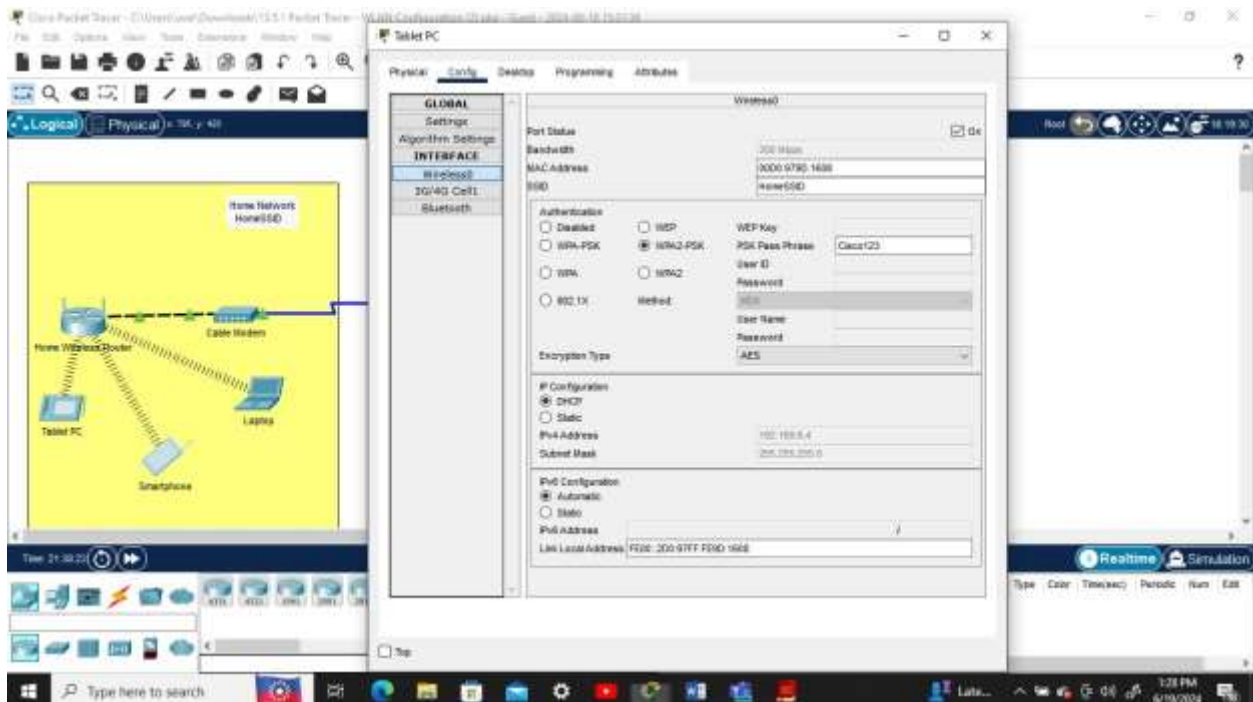
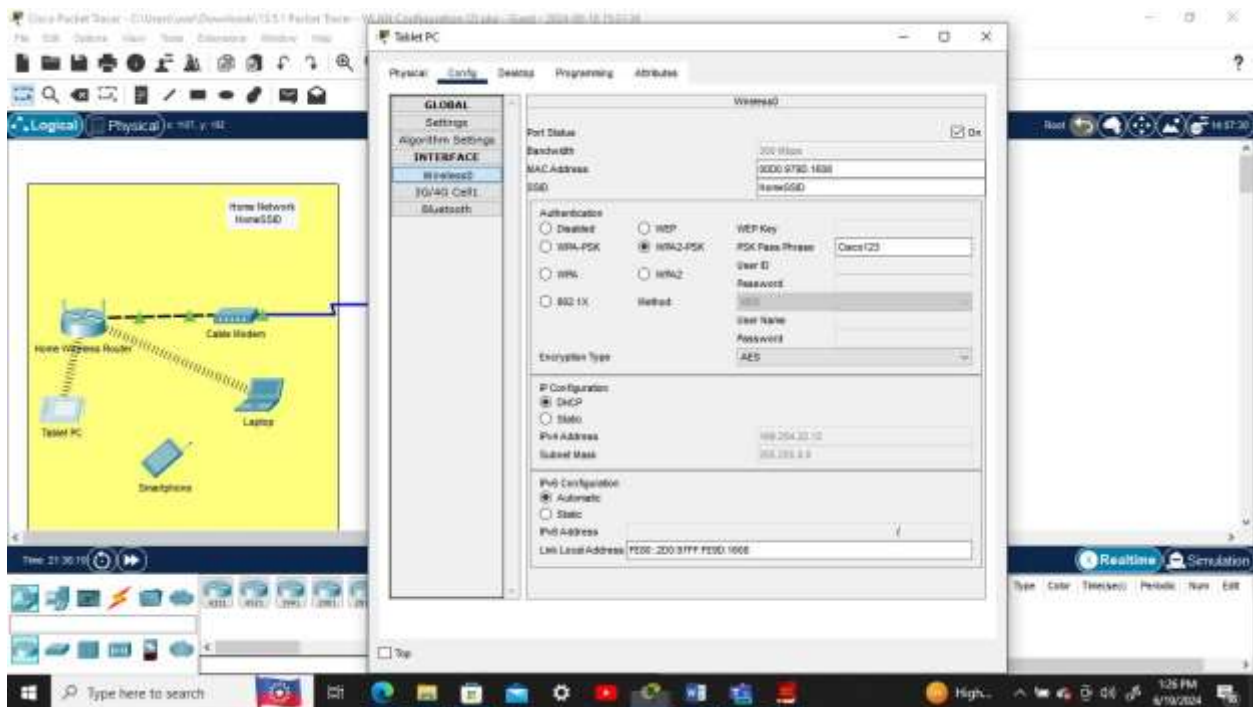


Step 4: Connect clients to the network

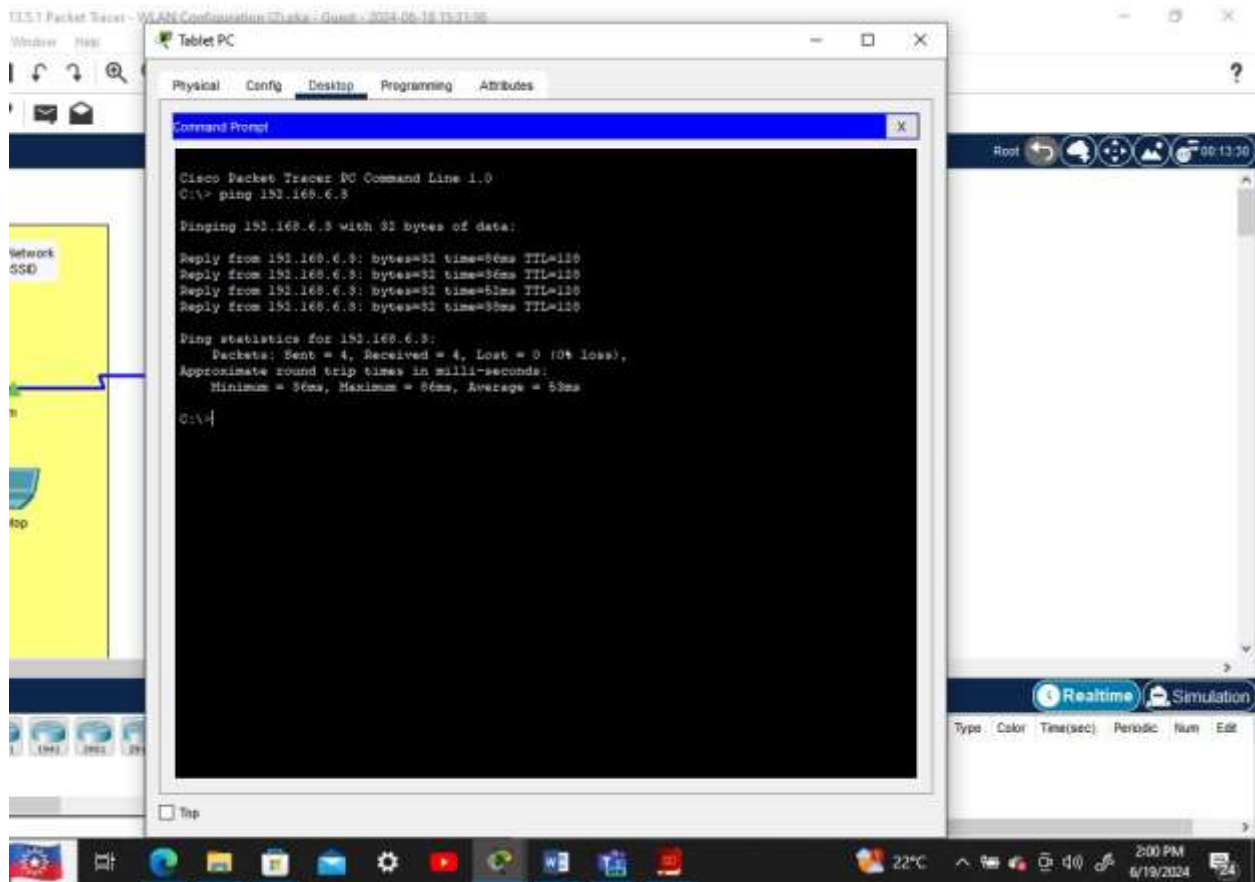
I opened the PC Wireless app on the desktop of the laptop and configured the client to connect to the network



After, I opened the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network



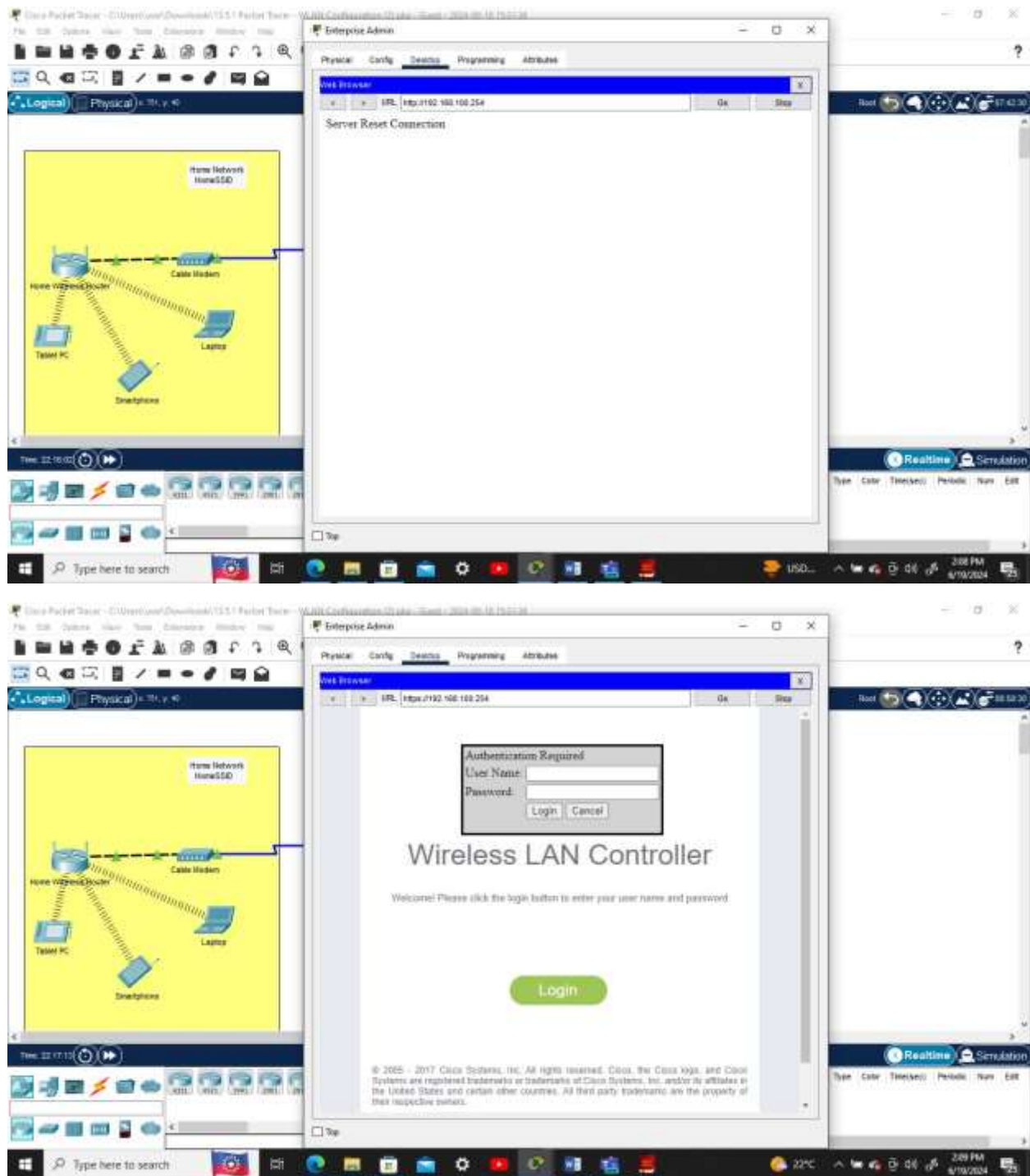
Next after that I verified connectivity. This was to ensure hosts could ping each other and the web server and that they could also reach the web server URL.



Part 2: Configure a WLC Controller Network

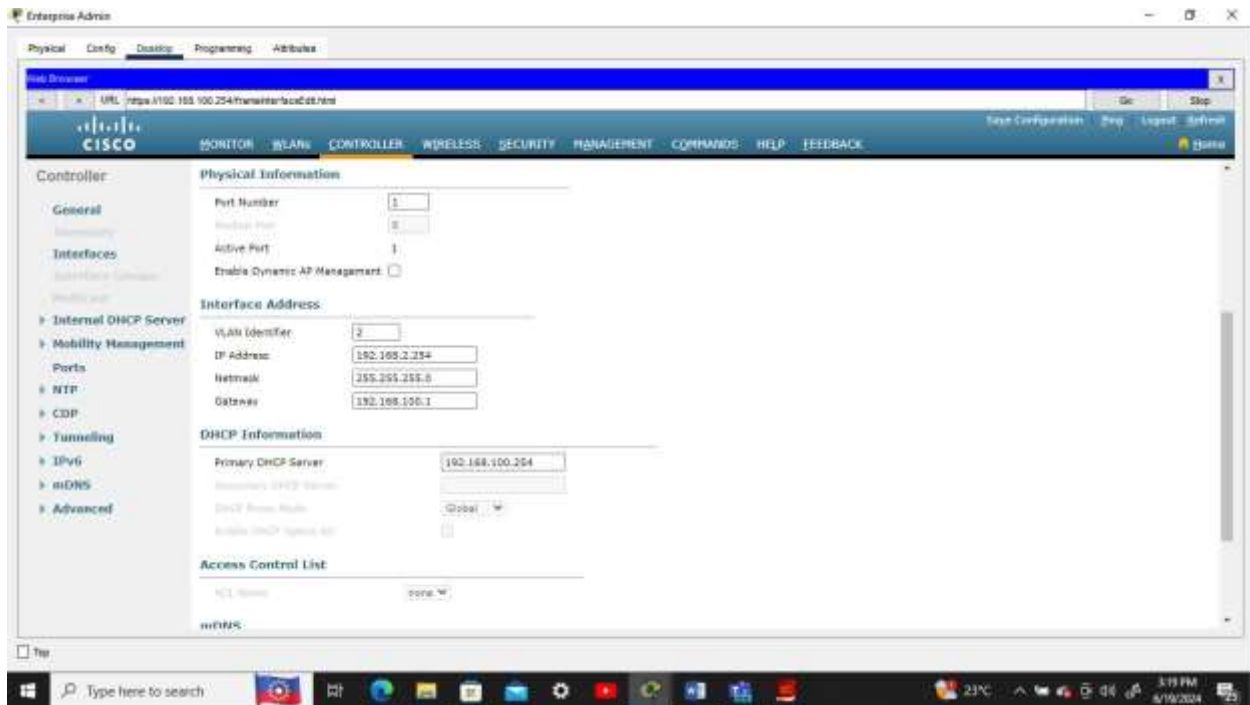
In this section, I configured the wireless LAN controller with two WLANs. One WLAN used the WPA2-PSK authentication and the other WLAN used WPA-2 Enterprise authentication. I also configured the WLC to use an SNMP server and configured a DHCP scope that was to be used by the wireless management network.

Step 1: Configure VLAN interfaces

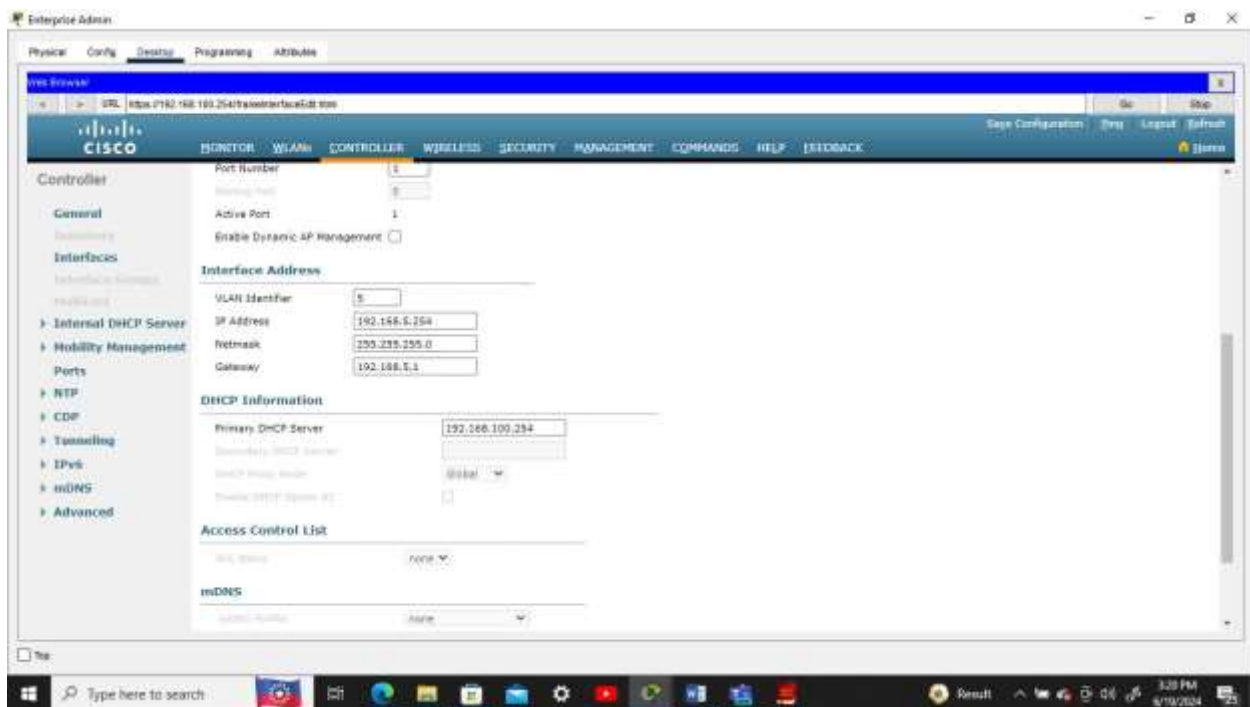


From an Enterprise Admin, I navigated to the WLC-1 management interface via a web browser. To gain access, I logged into WLC-1 and used **admin** as the username and **Cisco123** as the password

Step two was to configure an interface for the first WLAN



Step three was to configure an interface for the second WLAN

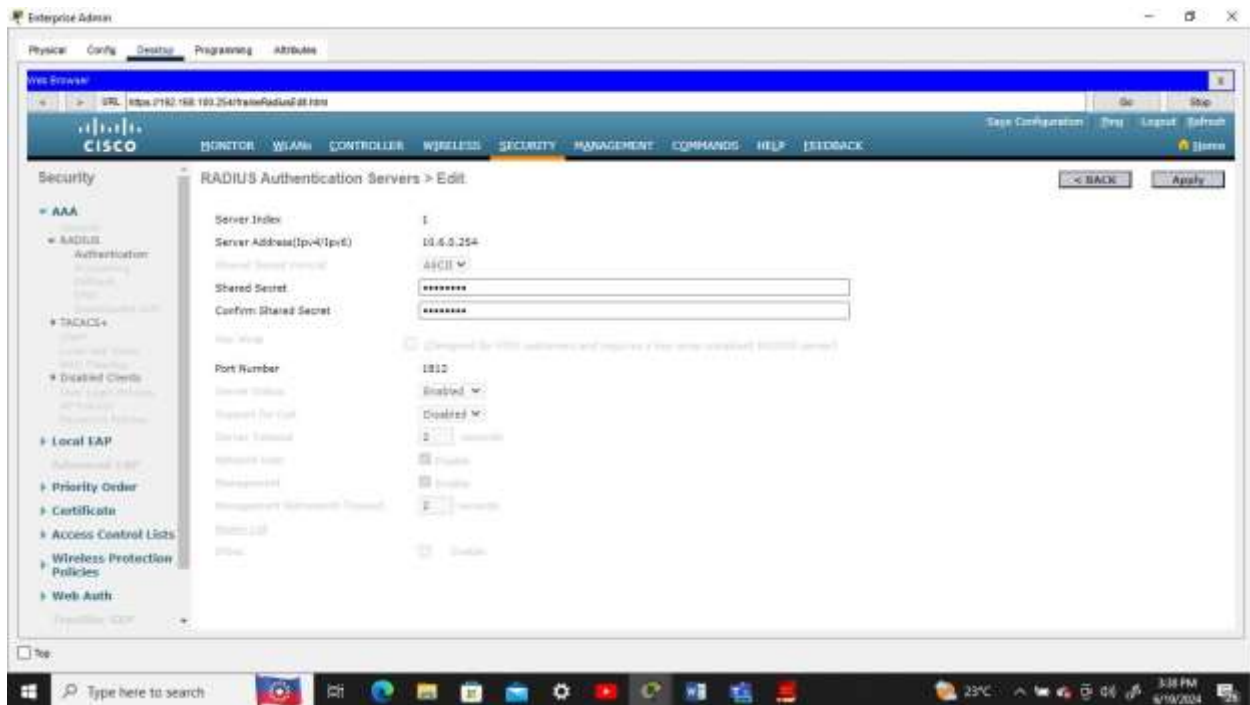


Step 2: Configure a DHCP scope for the wireless management network

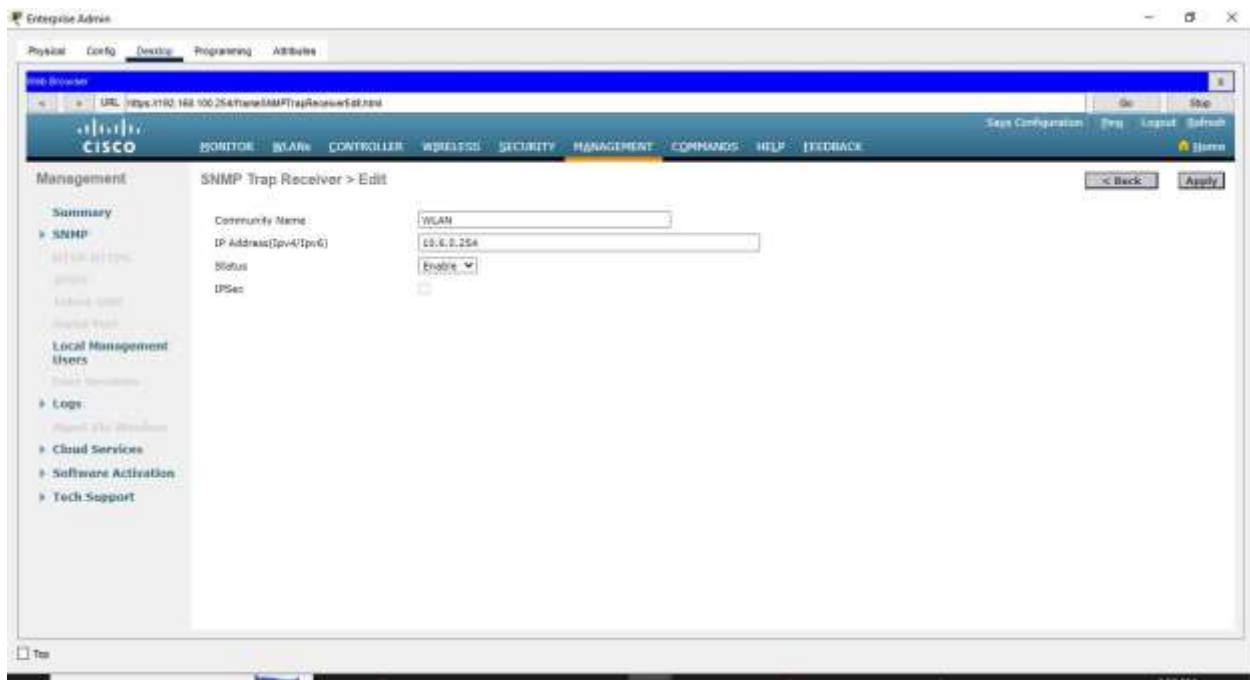
I configured and enabled an internal DHCP scope

Step 3: Configure the WLC with external server addresses

I configured the RADIUS server

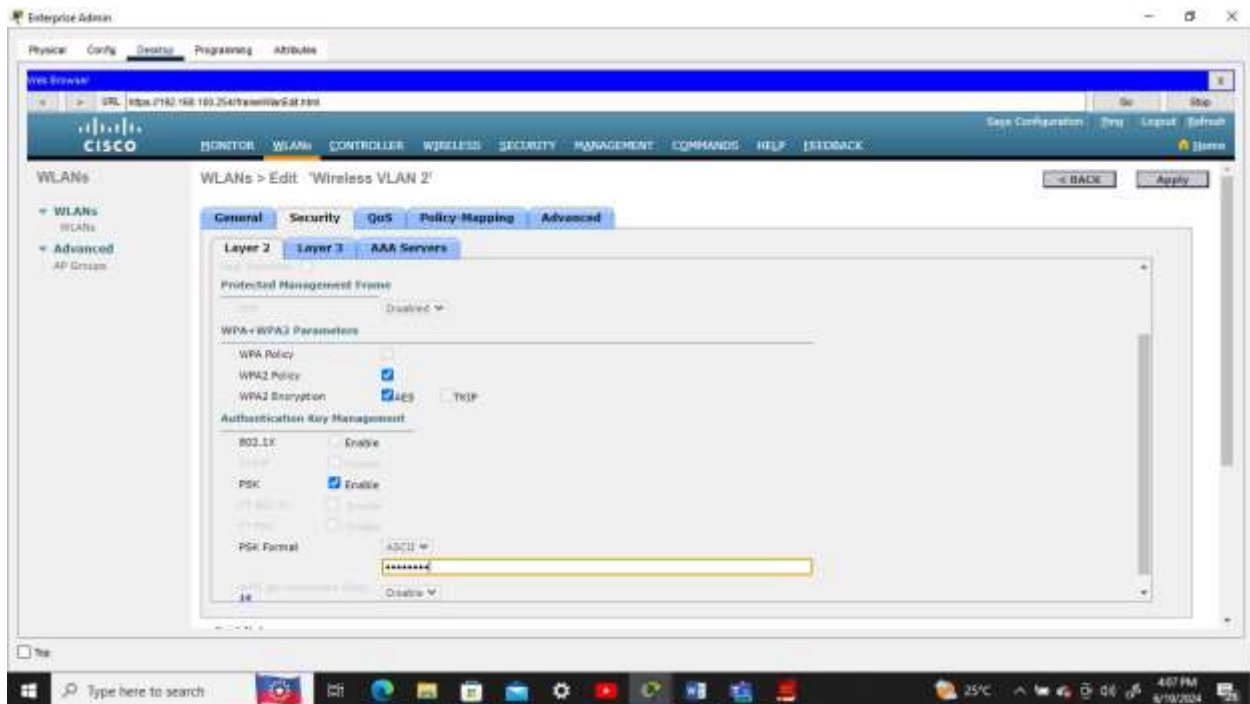
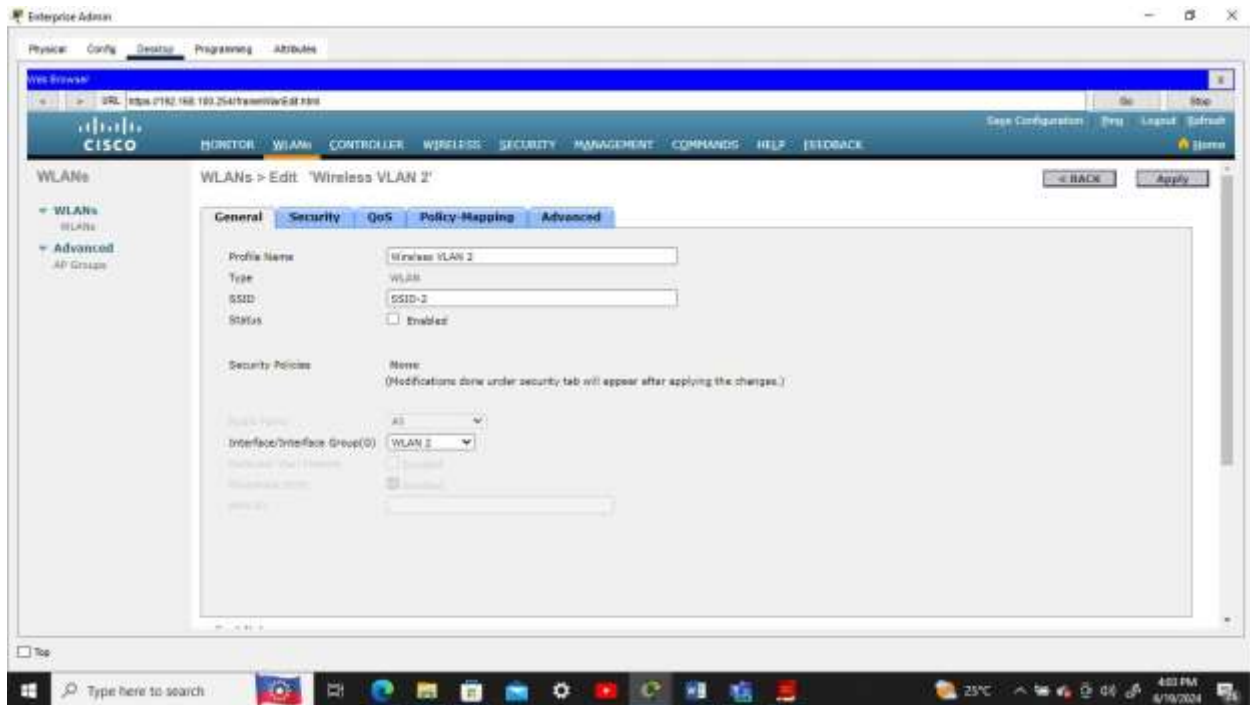


Then configured the WLC to send log information to an SNMP server.

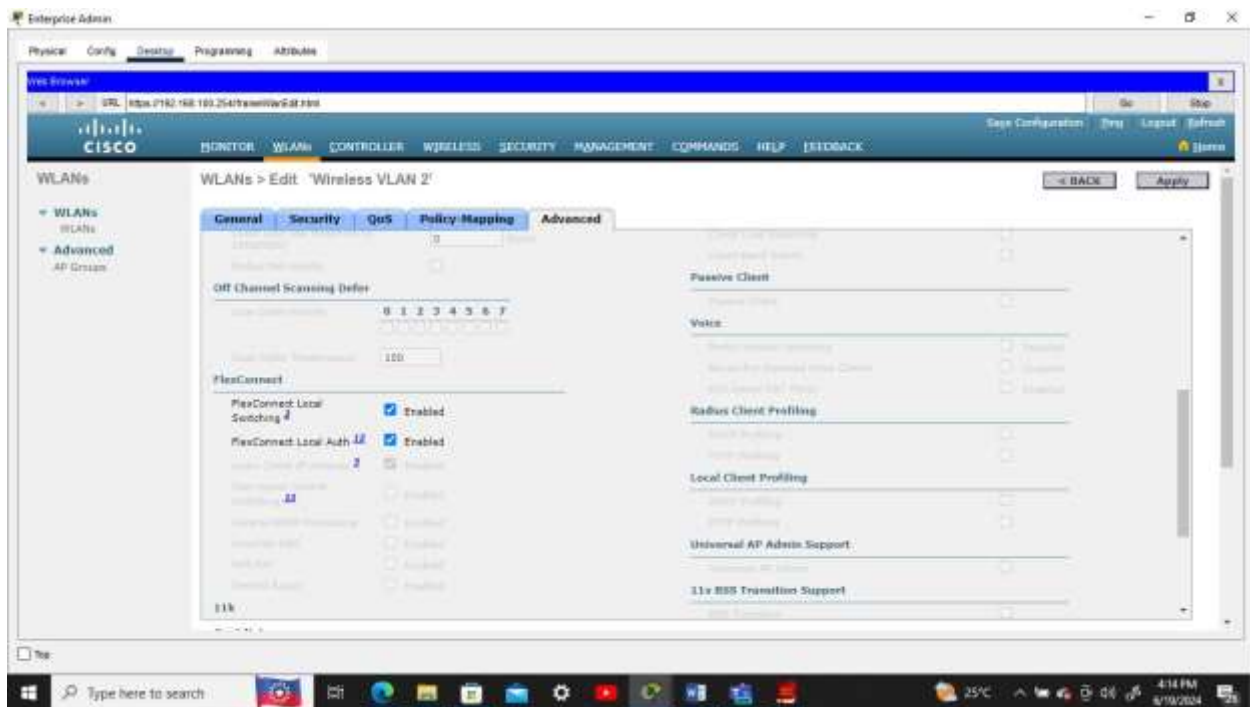


Step 4: Create the WLANs

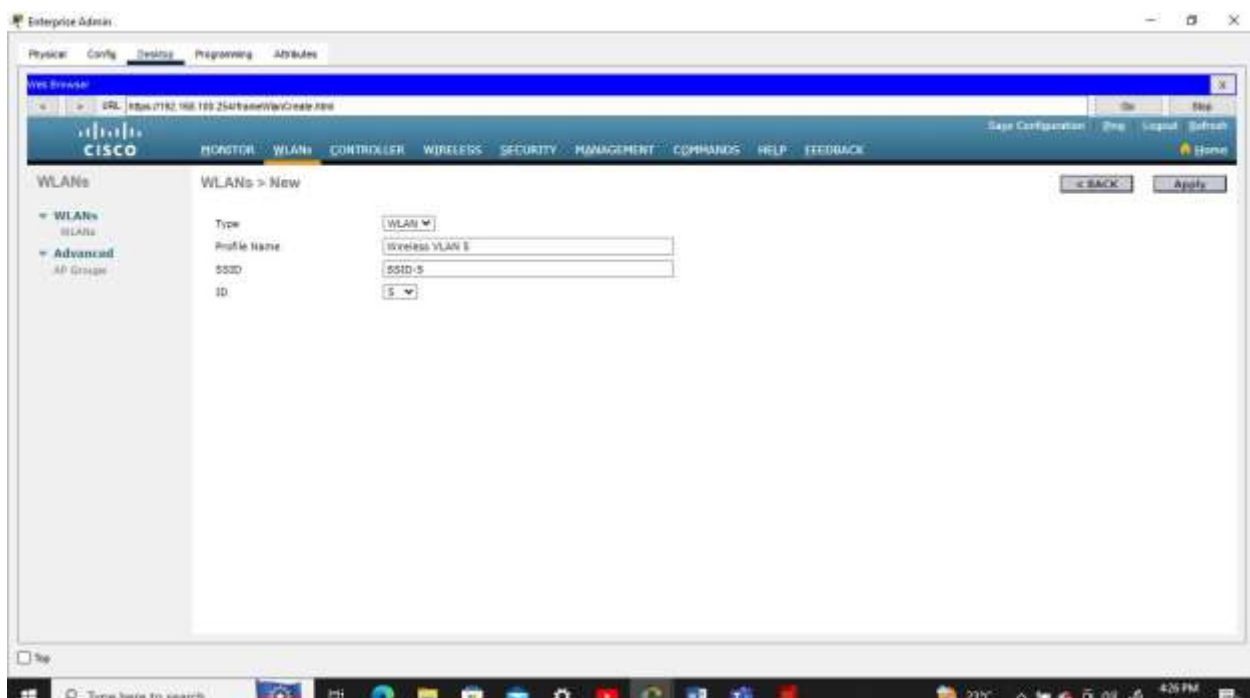
I created the first WLAN

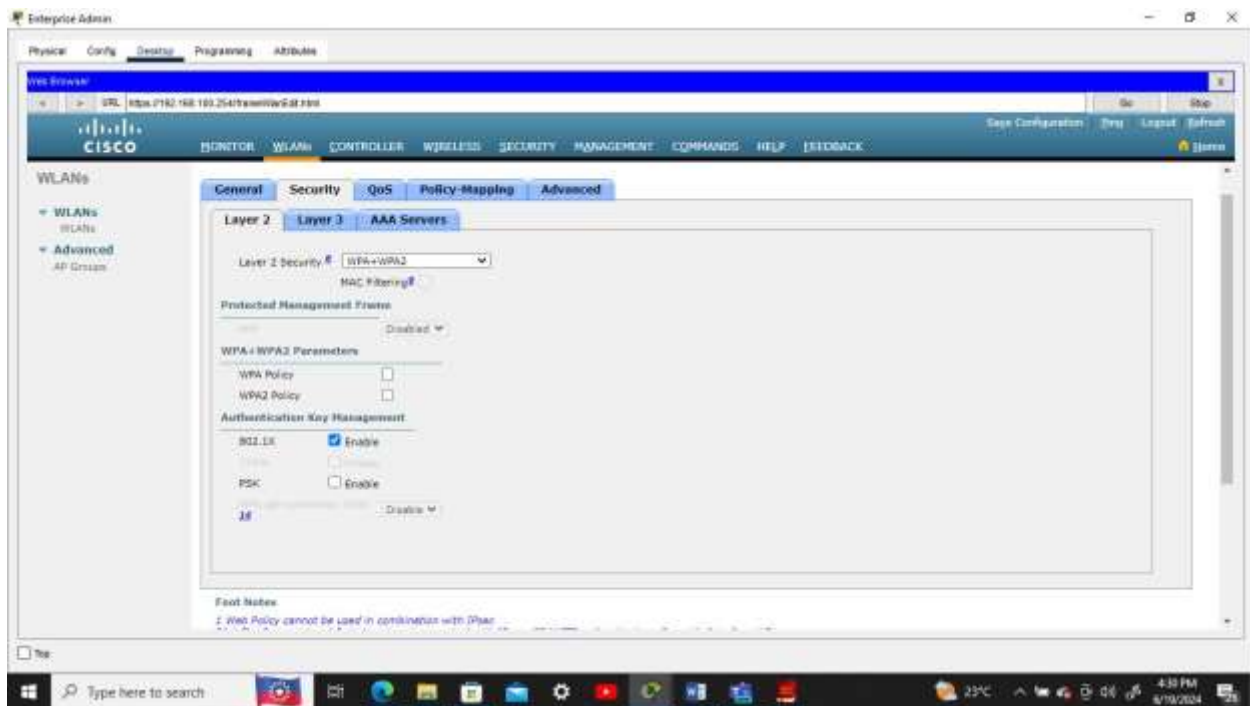
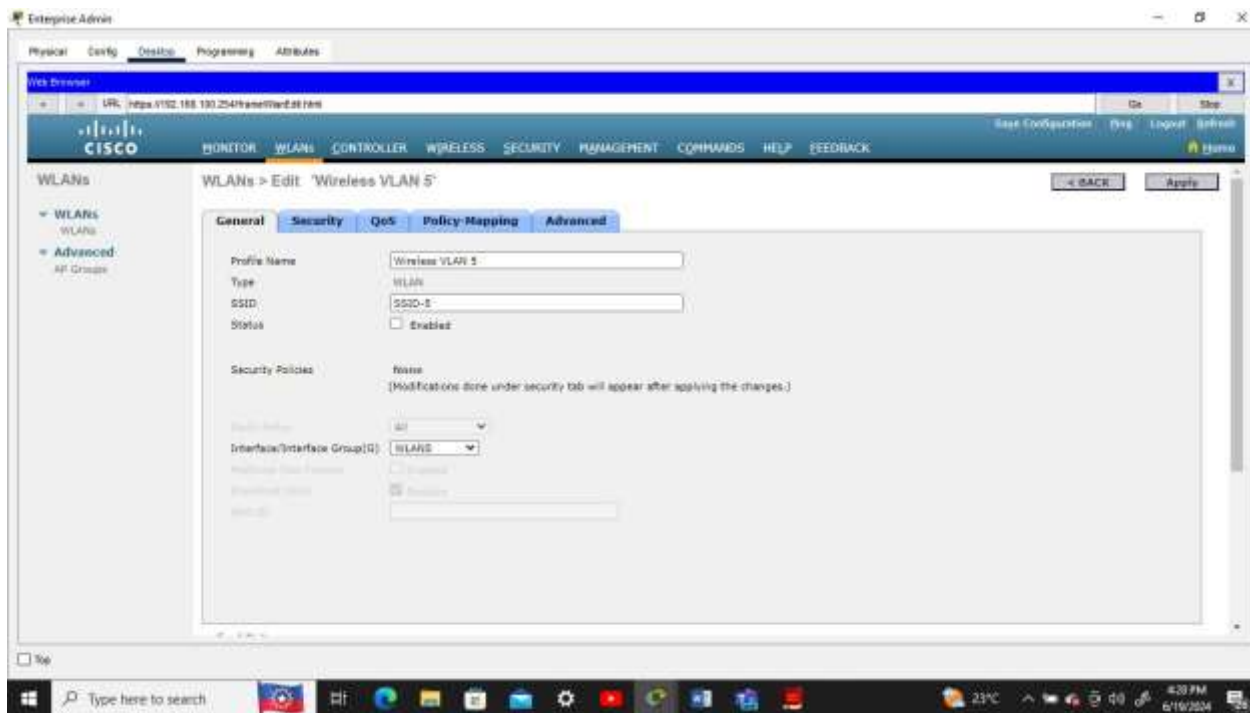


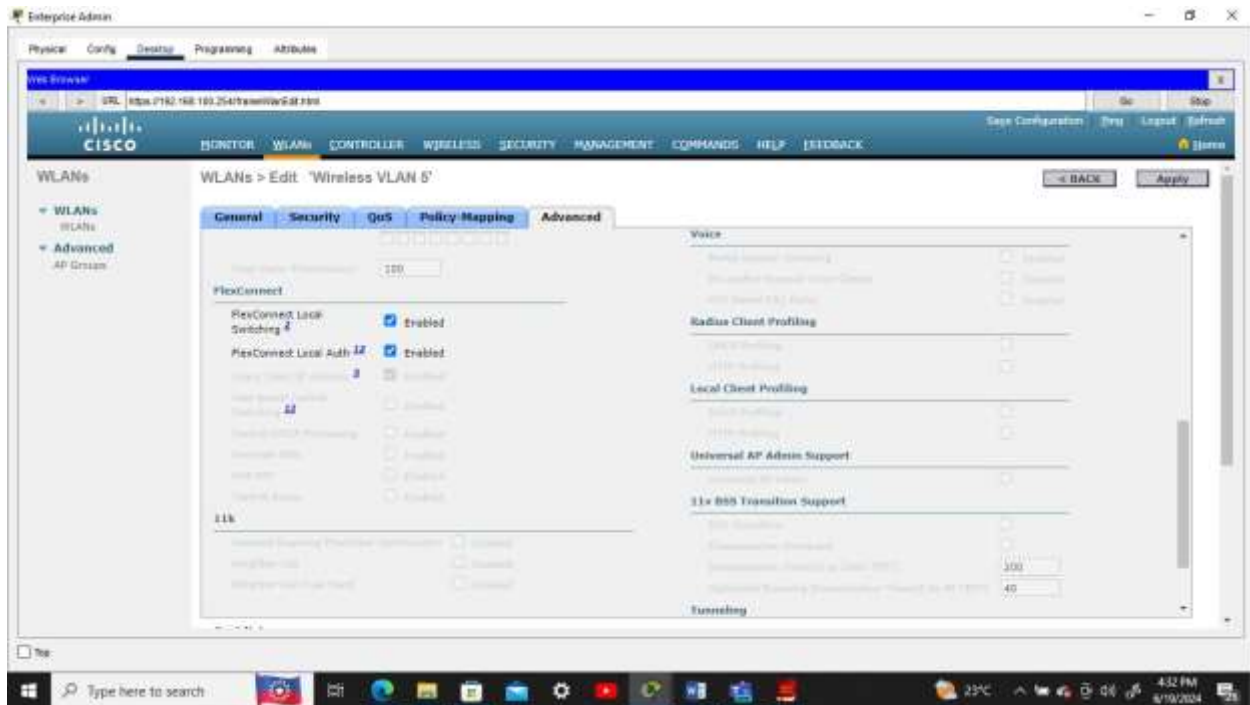
Under the Advanced tab, I went to the FLEXConnect section and enabled **FlexConnect Local Switching** and **FlexConnect Local Auth**.



I then created the second WLAN







Conclusion

This assignment was informative on how to secure a home network and on how to configure devices to ensure connectivity and maximum security. I also learnt that WPA2-PSK simplifies network security by using a single password shared among all users. It is also suitable for small networks like home offices. Next is WPA2-Enterprise which uses a centralized server to manage user credentials, which in turn offers more robust security features. It was fun tackling this assignment and seeing how the network devices in a WLAN can be individually configured.